

1 AREA TABLE — NET SF (BASE CASE)

ROOM / SPACE	NET SF	BASIS	STATUS · CONF.
FIVE TAGGED MAIN SPACES	39,003		
Championship Wrestling Arena (event floor)	22,000	phase2.yaml tag (concept sheet)	TAG · HIGH
Boys Locker Room	3,500	phase2.yaml tag	TAG · HIGH
Girls Locker Room (as tagged; D-013)	3,503	phase2.yaml tag; D-013 open	TAG · HIGH
Strength & Conditioning	6,000	phase2.yaml tag	TAG · HIGH
Cross-Training Mat Area	4,000	phase2.yaml tag	TAG · HIGH
EVENT BOWL	16,800		
Spectator seating (2,200 seats)	13,200	2,200 x 6.0 SF/seat (Loudoun 2014)	CITED · MEDIUM
Event / visitor locker rooms (DR3)	3,600	4 x (14 wrestlers + 4 staff) x 50 SF	ASSUMED · LOW
PUBLIC / LOBBY	3,550		
Concourse / hall of champions spine	2,750	25% of seats x 5 SF standing (T1004.5)	ASSUMED · LOW
Foyer / trophy lobby (hall of champions)	800	200 + 400 + 200 (Loudoun 2014)	CITED · MEDIUM
SUPPORT ROOMS (9 ON PLAN)			
Public restrooms (2 on plan)	3,350	fixtures (T2902.1) x 50 SF	CITED · MEDIUM
Concession stand	275	Loudoun kitchen proxy; D-011	CITED · LOW
First aid	120	Loudoun 2014; 1 room counted	CITED · MEDIUM
Administrative office	470	150 + 120 + 200 (Loudoun 2014)	CITED · LOW
Equipment storage (mats + floor covers)	700	Loudoun gym storage proxy	CITED · LOW
Equipment room	300	Loudoun equip./maint. proxy	CITED · LOW
Stage	TBD	no source: left out	TBD · n/a
Utility room	TBD	no source: left out	TBD · n/a
Mechanical (2 rooms on plan)	4,305	5% of gross (MIL-HDBK-1027/4A)	CITED · LOW
NET PROGRAM (excl. mechanical; TBD rooms left out)	64,568		
Mechanical = 5% of gross (row above)	4,305		
GROSS SF = 1.25 x (net + mechanical): walls, structure, circulation	86,091		

Gross-up 1.25: +25% for walls, structure and circulation (Groton MA feasibility study); Loudoun County 2014 rec-center standard uses 80% efficiency for program space (÷0.80 = x1.25). Mechanical 5% of gross: MIL-HDBK-1027/4A (HVAC rooms). Formula: gross = 1.25 x net ÷ (1 – 1.25 x 0.05).

LEAN CHECK (most favorable cited inputs): telescopic-bleacher seating 3.38 SF/seat and gross-up 1.15 (Loudoun walls/structure allowance only) give 58,806 net -> 71,753 GSF. Still 16,753 SF over.

2 SEATING — 2,200 SEATS (R-008)

BASE: 6.0 SF/seat incl. aisles — Loudoun County 2014 recreation-center standard (spectator seating, elevated). 2,200 x 6.0 = 13,200 SF.

LEAN: telescopic bleachers 18 in/person (IBC 1004.6) x 24 in rows (Hussey MAXAM) = 3.0 SF, + one 48 in stepped aisle (IBC 1030.9.1) per 21 seats (1030.13.2.1) = 3.38 SF/seat -> 7,438 SF. No front walkway, cross aisles or wheelchair spaces counted.

Seating type is OPEN (D-009). Seats stay outside the event floor (locked). IBC 2021 assembly egress is Section 1030 (1029 in the 2018 edition).

3 OCCUPANT LOAD + FIXTURES (R-009)

Load = 2,200 seats (IBC 1004.6) + event floor 22,000 SF ÷ 50 gross (T1004.5 'exercise rooms', function ASSUMED) = 440 -> 2,640 total; half each sex = 1,320 (2902.1.1). IBC 2021 Table 2902.1, arenas (indoor sporting events):

Water closets — men	1/75 first 1,500	18
urinals may be up to 12 (IPC 424.2, 67%)		
Water closets — women	1/40 first 1,520	33
Lavatories — men	1/200	7
Lavatories — women	1/150	9
Drinking fountains	1/1,000 (total)	3
in concourse; no SF		
Service sink	1	1
utility room (TBD)		

Room fixtures (WC + lav) = 67 x 50 SF/fixture (ASSUMED; bare WC module from cited minimums ≈ 27 SF) = 3,350 SF. Athletes' toilets assumed inside the locker-room tags. Code edition: Shane said IBC/IPC 2021; Madison County's page lists 2018 — same Table 2902.1 ratios. AHJ OPEN (D-008).

4 RESULT

DOES NOT FIT IN 55,000 SF

Needs ≈ 86,100 GSF (base) · ≈ 71,800 (lean)

Over by ≈ 31,100 SF (base) · ≈ 16,800 SF (lean)

The 5 tagged spaces alone are 39,003 net SF; grossed up they need ≈ 52,000 GSF before any seats, lockers or lobby.

5 OPTIONS (SF MATH)

A GROW THE BUILDING
to ≈ 86,100 GSF (base) or ≈ 71,800 GSF (lean); keeps all 2,200 seats and the 22,000 SF floor.

B SHRINK THE EVENT FLOOR
to 114 x 126 ft = 14,364 SF: 4 mats 42 ft, 2x2, 10 ft around and between (NFHS 2-1-5), 6 ft table strips N + S (ASSUMED); a 84x50 court + 10 ft runoff (104x70) fits inside. Building ≈ 75,800 GSF (base) / ≈ 62,300 (lean): still over by ≈ 20,800 / 7,300.

C CUT SEATS
not enough alone: with 0 seats the building is still ≈ 61,200 GSF (base) / ≈ 56,000 (lean). With B + C: ≈ 360 seats fit (base) / ≈ 1,190 (lean).

Not credited: a 2nd level, a mezzanine, or rooms under the seating bowl (DR3). Whether 55,000 SF caps total floor area or footprint is OPEN (D-030).

- 6 SOURCES (retrieved 2026-10-03)**
- IBC 2021 (UpCodes, AL ed.): T1004.5, 1004.6, 1030.9.1, 1030.13.2.1, T2902.1, 2902.1.1; IPC 2021 424.2; IBC 2018 T2902.1 (compared)
 - Madison County AL, Building Codes page (2018 I-codes)
 - NFHS Wrestling Rules 2-1-5, 2-3 (2014-15 text; current book paywalled) — R-005
 - NFHS Basketball Rules 1-1, 1-2-1 — R-012
 - AHSAA 2026-27 Wrestling (14 weight classes)
 - Loudoun County VA Capital Facilities Manual 2014, §4.3 Rec Center — R-014
 - Groton MA Senior Center feasibility study (25% net-to-gross) — R-014
 - MIL-HDBK-1027/4A in UFC 4-171-01N (HVAC 5% of gross) — R-014
 - Hussey Seating MAXAM brochure (row spacing); ASI restroom guide (stall sizes)

PRELIMINARY — CONCEPTUAL DESIGN — NOT FOR CONSTRUCTION

PROJECT	PHASE	SHEET TITLE	SCALE	DATE	REV	DRAWN BY	SHEET
Hazel Green Regional Athletic Complex Trojan Horse Arena · Hazel Green area, Madison County, Alabama (site TBD, D-006)	PHASE 2	PROGRAM TEST-FIT AREA ARITHMETIC, NOT A PLAN	NTS	October 2026	A	Drawn by KEYSTONE (AI) for Shane Brazelton	P2-G-003